VIVOCIN

(Vancomycin Hydrochloride for Injection USP 500mg/Vial)

DESCRIPTION AND COMPOSITION:

Description:
Before reconstitution: White to almost white lyophilized powder or cake

After reconstitution: A clear colorless solution, free from visible particulate matter.

This product needs to be reconstituted and then undergo further dilution before it can be administered

Reconstitution for 500mg: 500 milligrams vial should be reconstituted with 10mL of Water for Injections. The resulting solution contains Vancomycin 50 mg/mL.

For Further dilution: The resulting solutions containing 500 milligrams of Vancomycin must be further diluted with at least 100 mL of sodium chloride Intravenous Infusion 0.9% or Glucose Intravenous Infusion 5%.

Description after further dilution: A clear colorless solution, free from visible particulate matter, after dilution with 0.9% Sodium Chloride Solution & with 5 % Glucose solution.

Composition: Each vial contains Vancomycin Hydrochloride USP equivalent to Vancomycin 500 mg.

PHARMACODYNAMICS
General properties

ATC classification
Pharmacotherapeutic group: glycopeptide antibacterials, ATC code: J01XA01.

Mode of action: Vancomycin is a glycopeptide antibiotic. Vancomycin has a bactericidal effect on proliferating germs by inhibiting the biosynthesis of the cell wall. In addition, it impairs the permeability of the bacterial cell membrane and RNA synthesis.

Mechanism(s) of resistance: Acquired resistance to glycopeptides is based on acquisition of various van gene complexes. Van genes have rarely been found in Staphylococcus aureus, where changes in cell wall structure result in "intermediate" susceptibility, which is most commonly heterogeneous.
There is no cross-resistance between vancomycin and other antibiotics but cross-resistance with other glycopeptide antibiotics, such as teicoplanin, does occur. Secondary development of resistance during therapy is rare.
In some countries, increasing cases of resistance are observed particularly in enterococci; multi-resistant strains of Enterococcus faecium are especially alarming.

Synergism: The combination of vancomycin with an aminoglycoside antibiotic has a synergistic effect against many strains of Staphylococcus aureus, non-enterococcal D-streptococci, enterococci and streptococci of the Viridans group. The combination of vancomycin with a cephalosporin has a synergistic effect against some oxacillin-resistant Staphylococcus epidermidis strains, and the combination of vancomycin with rifampicin has a synergistic effect against Staphylococcus epidermidis and a partial synergistic effect against some Staphylococcus aureus strains. As vancomycin in combination with a cephalosporin may also have an antagonistic effect against some Staphylococcus epidermidis strains and in combination with rifampicin against some Staphylococcus aureus strains, preceding synergism testing is useful.
Specimens for bacterial cultures should be obtained in order to isolate and indentify the causative organisms and to determine their susceptibility to vancomycin.

Breakpoints: Minimum inhibitory concentration (MIC) breakpoint established by the European Committee on Antimicrobial Susceptibility Testing (EUCAST) for Staphylococcus spp. and Streptococcus spp.are Susceptible ≤ 2 mg/L and Resistant > 2 mg/L; forEnterococcus spp. are Susceptible ≤ 4 mg/L and Resistant > 4 mg/L; and for non-species related are Susceptible ≤2 mg/L and Resistant > 4 mg/L.

Susceptibility: The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent is at least some types of infections is questionable.

Vancomycin has a narrow spectrum of action.

Commonly susceptible species
Staphylococcus spp Streptococcus pneumonia Streptococcus spp Corynebacterium spp Enterococcus spp
Species for which acquired resistance may be a problem
Enterococcus faecium
inherently resistant organisms
Gram-negative bacteria, mycobacteria, fungi

PHARMACOKINETICS
Distribution: Following intravenous administration, vancomycin is distributed to almost all tissues and diffuses in pleural, pericardial, ascitic and synovial fluid as well as in the cardiac muscle and in heart valves. Comparable high concentrations are achieved as in blood plasma. Data about the vancomycin concentrations in bone (spongiosa, compacta) vary widely. The apparent distribution volume in steady state is stated to be 0.43 (up to 0.9) L/kg. In non-inflamed meninges vancomycin passes the blood-brain barrier only to a low extent. Vancomycin is bound to plasma proteins at 30 to 55 % and even higher.

Elimination: Vancomycin is metabolized only to a low extent. After parenteral administration it is excreted almost completely as microbiologically active substance (approx. 75-90% within 24 hours) through glomerular filtration via the kidneys. Biliary excretion is insignificant (less than 5% of a dose).
In patients with normal renal function the half-life in serum is about 4-6 (5-11) hours, in children 2.2-3 hours. In impaired renal function, the half-life of vancomycin may be considerably prolonged (up to 7.5 days). Due to ototoxicity of vancomycin therapy-adjuvant monitoring of the plasma concentrations is indicated in such cases.
Mean plasma concentrations after i.v. infusion of 1000 mg vancomycin over 60 minutes were about 63 mg/L at the end of the infusion, about 23 mg/L after 2 hours and about 8 mg/L after 11 hours.
The clearance of vancomycin from plasma correlates nearly with the glomerular filtration rate. The total systemic and renal clearance of vancomycin can be reduced in elderly patients.
As studies in anephric patients showed, the metabolic clearance seems to be very low. No vancomycin metabolites have been identified so far in humans.
If vancomycin is given during a peritoneal dialysis via the intraperitoneal route, approx. 60% reaches the systemic circulation during 6 hours. After i.p. administration of 30 mg/kg BW, serum levels of approx. 10 mg/l are achieved.
In case of oral use, high-polar vancomycin is virtually not absorbed. It appears after oral administration in active form in the stool, and is therefore a suitable chemotherapeutic for pseudomembranous colitis and staphylococcal colitis.
Vancomycin diffuses readily across the placenta and is distributed into cord blood.

INDICATIONS
Vancomycin is indicated for the treatment of serious or severe infections caused by susceptible strains of methicillin- resistant (beta-lactam-resistant) staphylococci. It is indicated for penicillin-allergic patients, for patients who cannot receive or who have failed to respond to other drugs, including the penicillins or cephalosporins, and for infections caused by vancomycin-susceptible organisms that are resistant to other antimicrobial drugs. Vancomycin is indicated for initial therapy when methicillin-resistant staphylococci are suspected, but after susceptibility data are available, therapy should be adjusted accordingly.
Vancomycin's effectiveness has been documented in other infections due to staphylococci, including septicemia, bone infections, lower respiratory tract infections, skin, and skin structure infections. When staphylococcal infections are localized and purulent, antibiotics are used as adjuncts to appropriate surgical measures.

Specimens for bacteriologic cultures should be obtained in order to isolate and identify causative organisms and to determine their susceptibilities to vancomycin.
To reduce the development of drug-resistant bacteria and maintain the effectiveness of vancomycin and other antibacterial drugs, vancomycin should be used only to treat or prevent infections that are proven or strongly suspected to be caused by susceptible bacteria. When culture and susceptibility information are available, they should be considered in selecting or modifying antibacterial therapy. In the absence of such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy.

RECOMMENDED DOSE
Vancomycin Hydrochloride for Injection, USP is intended for intravenous use. In selected patients in need of fluid restriction, a concentration up to 10 mg/mL may be used; use of such higher concentrations may increase the risk of infusion-related events. An infusion rate of 10 mg/min or less is associated with fewer infusion-related events (see ADVERSE REACTIONS). Infusion-related events are related to both concentration and rate of administration of vancomycin. Concentrations of no more than 5 mg/mL and rates of no more than 10 mg/min are recommended in adults (see also age-specific recommendations).

Patients with Normal Renal Function
Adults: The usual daily intravenous dose is 2 g divided either as 500 mg every six hours or 1 g every 12 hours. Each dose should be administered at no more than 10 mg/min, or over a period of at least 60 minutes, whichever is longer. Other patient factors, such as age or obesity, may call for modification of the usual intravenous daily dose.

Pediatric Patients: The usual intravenous dosage of vancomycin is 10 mg/kg per dose given every six hours. Each dose should be administered over a period of at least 60 minutes. Close monitoring of serum concentrations of vancomycin is recommended in these patients.

Neonates: In pediatric patients up to the age of 1 month, the total daily intravenous dosage may be lower. In neonates, an initial dose of 15 mg/kg is suggested, followed by 10 mg/kg every 12 hours for neonates in the first week of life and every eight hours thereafter up to the age of one month. Each dose should be administered over 60 minutes. In premature infants, vancomycin clearance decreases as postconceptional age decreases. Therefore, longer dosing intervals may be necessary in premature infants. Close monitoring of serum concentrations of vancomycin is recommended in these patients.

Patients with Impaired Renal Function and Elderly Patients
Dosage adjustment must be made in patients with impaired renal function. In the elderly, greater dosage reductions than expected may be necessary because of decreased renal function. Measurement of vancomycin serum concentrations can be helpful in optimizing therapy, especially in seriously ill patients with changing renal function. Vancomycin serum concentrations can be determined by use of microbiologic assay, radioimmunoassay, fluorescence polarization immunoassay, fluorescence immunoassay, or high-pressure liquid chromatography.
If creatinine clearance can be measured or estimated accurately, the dosage for most patients with renal impairment can be calculated using the following table. The dosage of vancomycin per day in mg is about 15 times the glomerular filtration rate in mL/min:

DOSAGE TABLE FOR VANCOMYCIN
IN PATIENTS WITH IMPAIRED RENAL FUNCTION
(Adapted from Moellering et al)3

Creatinine Clearance mL/min	Vancomycin Dose mg/24 h
100	1545
90	1390
80	1235
70	1080
60	925
50	770
40	620
30	465
20	310
10	155

The initial dose should be no less than 15 mg/kg, even in patients with mild to moderate renal insufficiency.
The table is not valid for functionally anephric patients. For such patients, an initial dose of 15 mg/kg of body weight should be given to achieve prompt therapeutic serum concentrations. The dose required to maintain stable concentrations is 1.9 mg/kg/24 h. In patients with marked renal impairment, it may be more convenient to give maintenance doses of 250 to 1000 mg once every several days rather than administering the drug on a daily basis. In anuria, a dose of 1000 mg every 7 to 10 days has been recommended.

When only the serum creatinine concentration is known, the following formula (based on sex, weight, and age of the patient) may be used to calculate creatinine clearance. Calculated creatinine clearances (mL/min) are only estimates. The creatinine clearance should be measured promptly.

$$\text{Men: } \frac{\text{Weight (kg) x (140 - age in years)}}{\text{72 x serum creatinine concentration (mg/dL)}}$$

Women: 0.85 x above value
The serum creatinine must represent a steady state of renal function. Otherwise the estimated value for creatinine clearance is not valid. Such a calculated clearance is an overestimate of actual clearance in patients with conditions:
(1) characterized by decreasing renal function, such as shock, severe heart failure, or oliguria;
(2) in which a normal relationship between muscle mass and total body weight is not present, such as obese patients or those with liver disease, edema, or ascites; and (3) accompanied by debilitation, malnutrition, or inactivity.
The safety and efficacy of vancomycin administration by the intrathecal (intralumbar or intraventricular) routes have not been established.
Intermittent infusion is the recommended method of administration.

Instructions for Use:
For Intravenous Administration:
At the time of use, the 500milligrams vial should be reconstituted with 10mL of Water for Injections. The resulting solution contains vancomycin 50 mg/mL. The reconstituted solutions containing 500 milligrams of vancomycin must be further diluted with at least 100 mL of sodium chloride Intravenous Infusion 0.9% or Glucose Intravenous Infusion 5%. The desired dose, diluted in this manner, should be administered by intermittent intravenous infusion over a period of at least 60 minutes. Prior to administration, parenteral drug products should be inspected visually for particulate matter and discoloration whenever solution and container permit. After reconstitution, the vials have to be used immediately.

MODE OF ADMINISTRATION
Parenteral-Intravenous

CONTRAINDICATION
Vancomycin Hydrochloride for Injection is contraindicated in patients with known hypersensitivity to Vancomycin Hydrochloride.

WARNING AND PRECAUTIONS
Warnings:
In the presence of acute anuria or cochlear damage, vancomycin must be used only when absolutely necessary and if no other safer alternatives are available.
In case of severe acute hypersensitivity reactions (e.g. anaphylaxis), the treatment with vancomycin has to be discontinued immediately and the usual appropriate emergency measures have to be started (e.g. antihistaminics, corticosteroides, and – if necessary – artificial respiration).

Rapid bolus administration (i.e. over several minutes) may be associated with severe hypotension (including shock and rare cardiac arrest), histamine like responses and maculopapular or erythematous rash ("red man's syndrome" or "red neck syndrome"). Vancomycin should be infused slowly in a dilute solution (2.5 to 5.0 g/l) at a rate no greater than 10 mg/min and over a period not less than 60 minutes to avoid rapid infusion-related reactions. Stopping the infusion usually results in a prompt cessation of these reactions.
Vancomycin must be administered only by intravenous use, owing to the risk of necrosis. The risk of venous irritation is minimized by giving vancomycin in the form of a dilute infusion and by changing the injection site.
The administration of vancomycin by intraperitoneal injection during continuous ambulatory peritoneal dialysis has been associated with a syndrome of chemical peritonitis.
Nephrotoxicity: vancomycin must be used with caution in patients with renal failure as the possibility of developing toxic effects is much higher in the presence of prolonged high blood concentrations. In the treatment of these patients and in those who are receiving concomitant treatment with other nephrotoxic active substances (i.e. aminoglycosides), serial tests of renal function must be performed and the appropriate dose regimens adhered to in order to reduce the risk of nephrotoxicity to a minimum.
Ototoxicity: Ototoxicity, which may be transitory or permanent has been reported in patients with prior deafness, who have received excessive intravenous doses, or who receive concomitant treatment with another ototoxic active substance such as an aminoglycoside. Deafness may be preceded by tinnitus. Experience with other antibiotics suggests that deafness may be progressive despite cessation of treatment. To reduce the risk of ototoxicity, blood levels should be determined periodically and periodic testing of auditory function is recommended.

Precautions:
Vancomycin is very irritating to tissue and causes injection site necrosis if injected intramuscularly. Pain and thrombophlebitis may occur in many patients receiving vancomycin and are occasionally severe. The frequency and severity of thrombophlebitis can be minimized by administering the medicinal product slowly as a dilute solution and by changing the sites of infusion regularly. The frequency of infusion-related reactions (hypotension, flushing, erythema, urticaria and pruritus) increases with the concomitant administration of anaesthetic agents. This may be reduced by administering the vancomycin by infusion over 60 minutes, before anaesthetic induction.
Vancomycin should be used with caution in patients with allergic reactions to teicoplanin, since crossed hypersensitivity reactions between vancomycin and teicoplanin have been reported.
Anaesthetic induced myocardial depression may be enhanced by vancomycin. During anaesthesia, doses must be well diluted and administered slowly with close cardiac monitoring. Position changes should be delayed until the infusion is completed to allow for postural adjustment.

In patients receiving vancomycin over a longer-term period or concurrently with other medications which may cause neutropenia or agranulocytosis, the leukocyte count should be monitored at regular intervals.
All patients receiving vancomycin should have periodic haematologic studies, urine analysis, liver and renal function tests.
Prolonged use of vancomycin may lead to superinfections with resistant microorganisms, therefore such patients should be regulatory monitored. If superinfection occurs during therapy, appropriate measures should be taken.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including vancomycin, and may range in the severity from mild to life-threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of vancomycin. Antiperistaltics are contraindicated.

Regular monitoring of the blood levels of vancomycin is indicated in longer-term use, particularly in patients with renal dysfunction or impaired faculty of hearing as well as in concurrent administration of nephrotoxic or ototoxic substances, respectively.
Doses should be titrated on the basis of serum levels. Blood levels should be monitored and renal function tests performed regularly.

It is a general recommendation to monitor the concentrations 2-3 times weekly.

The elderly are particularly susceptible to auditory damage and should be given serial tests for auditory function if over the age of 60. Concurrent or sequential use of other neurotoxic substances should be avoided.

Vancomycin should be used with particular care in premature infants and children, because of their renal immaturity and the possible increase in the serum concentration of vancomycin. The blood concentrations of vancomycin should therefore be monitored carefully. The concomitant use of vancomycin and anaesthetic agents in children has been associated with erythema and anaphylactoid reactions. If the administration of vancomycin is required for surgical prophylaxis, it is advisable to administer the anaesthetic agents after completion of the vancomycin infusion.

INTERACTIONS WITH OTHER MEDICAMENTS
Other potentially nephrotoxic or ototoxic medications
Concurrent or sequential administration of vancomycin with other potentially neurotoxic or/and nephrotoxic active substances particularly gentamycin, amphotericin B, streptomycin, neomycin, kanamycin, amikacin, tobramycin, viomycin, bacitracin, polymyxin B, colistin and cisplatin may potentiate the nephrotoxicity and/or ototoxicity of vancomycin and consequently requires careful monitoring of the patient.
Because of synergic action (e.g. with gentamycin) in these cases the maximum dose of vancomycin has to be restricted to 500 mg every 8 hours.

Anesthetics
Concurrent administration of vancomycin and anaesthetic agents has been associated with erythema, histamine like flushing and anaphylactoid reactions. This may be reduced if the vancomycin is administered over 60 minutes before anesthetic induction.

Muscle relaxants
If vancomycin is administered during or directly after surgery, the effect (neuromuscular blockade) of muscle relaxants (such as succinylcholine) concurrently used can be enhanced and prolonged.

Incompatibilities
Vancomycin solutions have a low pH value. This may lead to chemical or physical instability if mixed with other substances. Therefore, each parenteral solution should be checked visually for precipitations and discolouration prior to use.
This medicinal product must not be mixed with other medicinal products except those mentioned in Product Description part.

Combination therapy
In case of combination therapy of vancomycin with other antibiotics/chemotherapeutics, the preparations should be administered separately.

Mixtures of solutions of vancomycin and beta-lactam antibiotics have been shown to be physically incompatible. The likelihood of precipitation increases with higher concentrations of vancomycin. It is recommended to adequately flush the intravenous lines between administrations of these antibiotics. It is also recommended to dilute solutions of vancomycin to 5 mg/mL or less.

PREGNANCYAND LACTATION
Pregnancy:
No sufficient safety experience is available regarding vancomycin during human pregnancy. Reproduction toxicological studies on animals do not suggest any effects on the development of the embryo, foetus or gestation period.

However, vancomycin penetrates the placenta and a potential risk of embryonal and neonatal ototoxicity and nephrotoxicity cannot be excluded. Therefore vancomycin should be given in pregnancy only if clearly needed and after a careful risk/benefit evaluation.

Lactation:
Vancomycin is excreted in human milk and should be therefore used in lactation period only if other antibiotics have failed. Vancomycin should be cautiously given to breast-feeding mothers because of potential adverse reactions in the infant (disturbances in the intestinal flora with diarrhoea, colonisation with yeast-like fungi and possibly sensibilisation).
Considering the importance of this medicine for nursing mother, the decision should to stop breastfeeding should be considered.

SIDE EFFECTS
Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

The adverse reactions listed below is defined using the following MedDRAconvention and system organ class database:

The most common adverse reactions are phlebitis and pseudo-allergic reactions in connection with too rapid intravenous use of vancomycin.

Blood and the lymphatic system disorders
Rare: thrombocytopenia, neutropenia, agranulocytosis, eosinophilia.

Immune system disorders
Rare: anaphylactic reactions, hypersensitivity reactions.

Ear and labyrinth disorders
Uncommon: transient or permanent loss of hearing. Rare: tinnitus, dizziness.

Cardiac disorders
Very rare: cardiac arrest.

Vascular disorders
Common: decrease in blood pressure, thrombophlebitis.
Rare: vasculitis.

Respiratory, thoracic and medistinal disorders
Common: dyspnoea, stridor.

Gastrointestinal disorders
Rare: nausea.
Very rare: pseudomembranous enterocolitis.

Skin and subcutaneous tissue disorders
Common: exanthema and mucosal inflammation, pruritus, urticaria.
Very rare: exfoliative dermatitis, Stevens-Johnson syndrome, Lyell's syndrome, IgAinduced bullous dermatitis.

Renal and urinary disorders
Common: renal insufficiency manifested primarily by increased serum creatinine or serum urea concentrations.
Rare: interstitial nephritis, acute renal failure.

General disorders and administration site conditions
Common: redness of the upper body and the face, pain and spasm of the chest and back muscles. Rare: drug fever, shivering.
During or shortly after rapid infusion anaphylactic reactions may occur. The reactions abate when administration is stopped, generally between 20 minutes and 2 hours after having stopped administration.
Ototoxicity has primarily been reported in patients given high doses, or concomitant treatment with other ototoxic medicinal products, or with pre-existing reduction in kidney function or hearing."

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES
Vancomycin has no or negligible influence on the ability to drive and use machines.

SYMPTOMS AND TREATMENT OF OVER DOSE
Toxicity due to overdose has been reported. 500 mg IV to a child, 2 year of age, resulted in lethal intoxication. Administration of a total of 56 g during 10 days to an adult resulted in renal insufficiency. In certain high-risk conditions (e. g. in case of severe renal impairment) high serum levels and ototoxic- and nephrotoxic effects can occur.

Measures in case of overdose
- Aspecific antidote is not known.
- Symptomatic treatment while maintaining renal function is required.
- Vancomycin is poorly removed from the blood by haemodialysis or peritoneal dialysis. Haemofiltration or haemoperfusion with polysulfone resins have been used to reduce serum concentrations of vancomycin.

STORAGE CONDITION
Storage condition for powder: Store below 30°C. Protect from light.
Storage condition for reconstituted solution: 30°C or 2 – 8°C for the duration of 96 hours
Storage conditions for further diluted solution: Diluted solution should be stored at 2 - 8°C for 14 days

SHELF LIFE
Powder before reconstitution: 24 months
After Reconstitution: The shelf life of product after reconstitution is 96 hours at 2 – 8°C or 30°C.
After Further dilution: The Shelf life of the product after further dilution is 14 days at 2 – 8°C.

DOSAGE FORM AND PACKAGING AVAILABLE:
10 x 10 ml vials packed in an outer carton along with the package insert.

MANUFACTURED BY
GLAND PHARMA LIMITED,
Unit - II, Plot No. 42 to 52,
Sy. No. 166, 171, 172 & 177,
Phase-III, TSIIC, Pashamylaram Village,
Patancheru Mandal, Sangareddy District - 502307,
Telangana, India

Product Registration Holder/
Imported by:
UNIMED SDN BHD
No. 53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara,
52200, Kuala Lumpur, Malaysia.

Date of Revision: July 2019

LEA-XXXXXX-XX

PRODUCT NAME	Vivocin	**ARTWORKS No.**	LEA-XXXXXX-XX
COUNTRY	Malaysia	**SUPERSEDES No.**	NA
CUSTOMER	Unimed SDN BHD	**PANTONE SHADE No's**	Black
DIMENSIONS	Open Dimension: 260 x 240 ± 2 mm(L x W) Folding Dimension: 43 x 120 ± 2 mm(L x W)	**VARNISHING/LAMINATE**	NA
SUBSTRATE	Maplitho paper 60 gsm ± 10%	**MODE OF SUPPLY**	Bundles / Tray pack

Form No:GPL/PP/F007-00/07-14